EGRE 245 Engineering Programming
Fall 2025
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Lab Exercise #4: Basic Calculator

1 Purpose

This lab will give you practice with if-else statements, switch statements, and logical operators. You will also learn how to handle invalid input and enforce simple rules for your program.

2 Overview

- Write a C program to implement a calculator that: 1) Uses if-else to check conditions like divide-by-zero; 2) Uses a switch statement to choose between operations; 3) Uses logical operators (&&, ——) to enforce rules about valid input.
- Ask the user to enter two numbers (floating-point is allowed).
- Enforce this rule with a logical operator:
  - If both numbers are positive OR their sum is greater than 10, the program continues.
  - Otherwise, print an error message and stop.
- If the input is valid, display this menu:
  1. Add
  2. Subtract
  3. Multiply
  4. Divide
  5. Modulus (only valid if both numbers are integers)
- Ask the user to choose an operation (1–5).
- Use a switch statement to perform the chosen operation.
- Use if-else to handle errors such as divide by zero or invalid modulus on floats.
- Print the result or error message clearly.

3 Code Documentation

The very first lines in *all* of your lab project files this semester should be comments in the class header file (copy from header.c). Of course, the items in the square brackets [] in the comments should be filled in with the proper information.

In addition, the first output for all of your projects this semester, for the lab exercises, should be the name of the assignment, your name and your lab partners name, e.g:

```
printf("Lab #4 - Joe Student and Jane Doe\n");
```

The next line of output should be `printf("EGRE 245:  Introduction to Programming Using C, September 23\n");`

Please fill in the correct date.

4 Sample Run

Your output should match the sample run below.

```
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Enter first number: 33

Enter second number: -12

Choose operation:
1. Add
2. Subtract
3. Multiply
4. Divide
5. Modulus

Your choice: 5


Result: 33 % -12 = 9


...Program finished with exit code 0
Press ENTER to exit console.
```

```
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Enter first number: 25

Enter second number:
0

Choose operation:
1. Add
2. Subtract
3. Multiply
4. Divide
5. Modulus

Your choice: 5


Error: Modulus only valid for nonzero integers.


...Program finished with exit code 0
Press ENTER to exit console.
```

```
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Enter first number: -10

Enter second number: 8


Invalid input: numbers must be positive or sum must be greater than 10.


...Program finished with exit code 0
Press ENTER to exit console.
```

5 Deliverables

You should turn in a stand-alone, complete application program (your C source code, a single ".c" file) containing a `main()` function. All labs and projects this semester will be submitted in Gradescope (via a Canvas assignment).

Please name your file Lab4_FirstName_LastName.c.

Due date: The lab is due at the end of your lab period, either Tuesday, September 23rd, or Thursday, September 25th.